

Mini-Golf: Structs and Functions

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1 Structs and Enums

1.1 types.h

enum SurfaceType

SURF_GREEN, SURF_FAIRWAY, SURF_SAND, SURF_ICE, SURF_MUD, SURF_WATER, SURF_BOOST,
SURF_COUNT

enum BallState

BALL_AIM, BALL_CHARGE, BALL_ROLL, BALL_SUNK, BALL_OOB

enum GameStateId

GS_MENU, GS_PLAYING, GS_HOLE_DONE, GS_SCOREBOARD, GS_PAUSED

struct Ball

Vector2 pos; Vector2 vel; float radius; BallState state; int strokes; Vector2
last_safe_pos; bool hitWater; bool hitWall; bool hitCup; bool hitByPutter;

struct Wall

Rectangle rect; float bounce;

struct Zone

Rectangle area; SurfaceType type; Vector2 wind;

struct Hole

Wall walls[MAX_WALLS]; int wall_count; Zone zones[MAX_ZONES]; int zone_count; int number;
int par; Vector2 tee_pos; Vector2 cup_pos; float cup_radius; Rectangle bounds; Rectangle
drop_zone;

struct Game

GameStateId state; Camera2D camera; Ball ball; Hole hole; int hole_index; int
scores[MAX_HOLES];

1.2 input.h

struct InputState

Vector2 aim_direction; float aim_angle; Vector2 pointer_world; bool pointer_valid; bool
charge_pressed; bool charge_held; bool charge_released; bool cancel; bool confirm; bool
cam_zoom_delta;

struct InputSystem

float aim_angle; float right_held_time; float right_drag_distance; bool right_was_down;

1.3 level.h

```
struct Course
Hole holes[MAX_HOLES]; int hole_count; int current;
```

1.4 putter.h

```
enum PutterPhase
putter_idle, putter_backswing, putter_strike, putter_follow, putter_recover
```

```
struct Putter
PutterPhase phase; float t; float power; float launch_power; float aim_angle; float
offset; float dir; bool fired;
```

1.5 render.h

```
struct RenderState
Camera2D cam; float hole_time;
```

1.6 main.c

```
struct PlayerScore
char name[MAX_NAME_LENGTH]; int score;
```

enum ScreenState

SCREEN_MAIN_MENU, SCREEN_NAME_INPUT, SCREEN_GAME, SCREEN_TUTORIAL, SCREEN_LEADERBOARD,
SCREEN_ABOUT

struct GameApp

GameStateId state; Course course; Ball ball; Putter putter; InputSystem input_sys;
RenderState render; int scores[MAX_HOLES]; int best[MAX_HOLES]; int bestCourseScore; bool
newBestScore; float hole_done_t; float prev_speed; ScreenState screen; Texture2D
menuBackground, tutorialImage, aboutImage, windSprite; Music menuMusic, gameMusic; Sound
waterSound, wallSound, cupSound, putterSound; bool muted; char
playerName[MAX_NAME_LENGTH]; int playerNameLength; PlayerScore leaderboard[MAX_PLAYERS];
int leaderboardCount;

2 Functions

2.1 ball.c

```
void BallInit(Ball *ball, Vector2 tee_pos)
```

```
float BallSpeed(Ball *ball)
```

```
bool BallAtRest(Ball *ball)
```

```
void BallLaunch(Ball *ball, Vector2 dir, float power)
```

```
void BallComeToRest(Ball *ball)
```

```
void CheckCup(Ball *ball, Hole *hole)
```

```
void CheckHazards(Ball *ball, Hole *hole)
```

```
void DropBall(Ball *ball, Hole *hole)
```

```
void StepPhysics(Ball *ball, Hole *hole, float dt)
```

```
void BallUpdate(Ball *ball, Hole *hole, float dt)
```

2.2 physics.c

```
float FrictionOf(SurfaceType s)
```

```
SurfaceType SurfaceAt(Hole *hole, Vector2 p)
```

```
Vector2 WindAt(Hole *hole, Vector2 p)
```

```
void ApplyFriction(Ball *ball, float decel, float dt)
```

```
void ApplyWind(Ball *b, Vector2 wind, float dt)
```

```
void CheckWallCollision(Ball *b, Hole *h)
```

```
int CheckCupCollision(Ball *b, Hole *h)
```

2.3 input.c

```
Vector2 DirFromAngle(float angle)
```

```
void InputInit(InputSystem *sys)
```

```
InputState InputPoll(InputSystem *sys, Vector2 ball_pos, Camera2D cam, float dt)
```

2.4 putter_simple.c

```
void putter_init(Putter *p)
```

```
bool PutterIsCharging(const Putter *p)
```

```
float PutterPower(const Putter *p)
```

```
void putter_update(Putter *p, const InputState *in, Ball *ball, float dt)
```

```
void putter_draw(const Putter *p, const Ball *ball)
```

2.5 level.c

```
void AddZone(Hole *h, SurfaceType t, float x, float y, float w, float ht, Vector2  
wind)
```

```
bool Validate_hole(Hole *h, const char *path)
```

```
void Parseline(Hole *h, char *line, int line_no, const char *path)
```

```
bool LoadHoleFromFile(Hole *h, const char *path)
```

```
int course_load(Course *c, const char *dir)
```

```
bool course_advance(Course *c)
```

```
Hole *course_current(Course *c)
```

```
const char *score_name(int strokes, int par)
```

```
int Course_total(const int *scores, int hole_count)
```

```
int course_to_par(const Course *c, const int *scores)
```

2.6 render.c

```
void RenderInit(RenderState *r, Vector2 ballpos)
```

```
void RenderUpdateCamera(RenderState *r, const Ball *b, const Hole *h, const Putter  
*p, float dt)
```

```
void DrawCourse(const Hole *h)
```

```
void DrawWindZones(const Hole *h, Texture2D sprite, float t)
```

```
void DrawRail(Rectangle r)
```

```
void DrawRails(const Hole *h)
```

```
void DrawBall(const Ball *b)
```

```
void DrawAimGuide(const Ball *b, const Hole *h, float angle, float power)
```

```
void DrawHUD(const Hole *h, const Ball *b, const Putter *p, int hole_index, int
hole_count, int total)
```

2.7 main.c

```
void DrawFullscreenTexture(Texture2D texture)
```

```
bool DrawMenuButton(Rectangle rect, const char *text)
```

```
void StartHole(GameApp *g)
```

```
void StartGameMusic(GameApp *g)
```

```
void StartMenuMusic(GameApp *g)
```

```
int LoadBestScore(void)
```

```
void SaveBestScore(int score)
```

```
void LoadLeaderboard(GameApp *g)
```

```
void SaveLeaderboard(GameApp *g)
```

```
void SortLeaderboard(GameApp *g)
```

```
void UpdateLeaderboard(GameApp *g, int score)
```

```
void CheckFinalScore(GameApp *g)
```

```
void RestartGame(GameApp *g)
```

```
void ResetGameProgress(GameApp *g)
```

```
int DrawFinalScoreScreen(GameApp *g)
```

```
void GameInit(GameApp *g)
```

```
bool DrawMainMenu(GameApp *g)
```

```
void DrawTutorialScreen(GameApp *g)
```

```
void DrawAboutScreen(GameApp *g)
```

```
void DrawNameInputScreen(GameApp *g)
```

```
void DrawLeaderboardScreen(GameApp *g)
```

```
int main(void)
```